

■ training variants      ■ held-out variants

## Random



random held-out variants

## Positional



hold out one protein region  
(C-terminal half)

## Mechanistic



hold out hotspot or  
functional-switch residues

## Cross-gene



hold out whole genes,  
train on the others

## Low-depth



hold out shallow-sampling  
variants

## Compatibility



match a prior evaluation  
protocol